

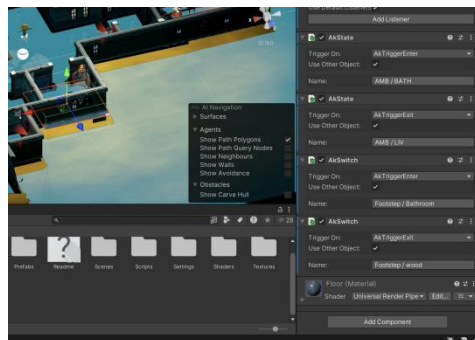
Work Description--John Lemon

Wwise Integration/Unity Game Project

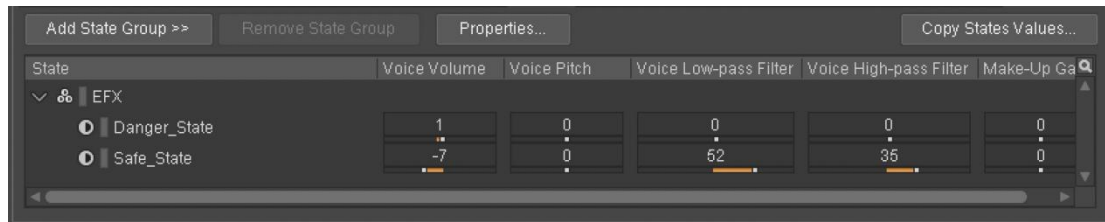
For this game sound design project, I chose the John Lemon demo from the Unity Asset store , which was inspired by one of my favorite horror-action games, Little Nightmares. My goal is to design a complete sound project, including composing a soundtrack for this game.

My first step is to organized a list of the necessary sounds:

- 1.Original music to fit the scene's atmosphere
- 2.Environmental sounds (such as rain from outside)
- 3.Two types of ghost sound (little ghost that moves around, gargoyle that holds a red flash light).
- 4.recorded footsteps with three different surface types (wood, carpet, bathroom tile) , the heartbeat of John lemon(main protagonist cat), and UI sound effects for both winning and losing.

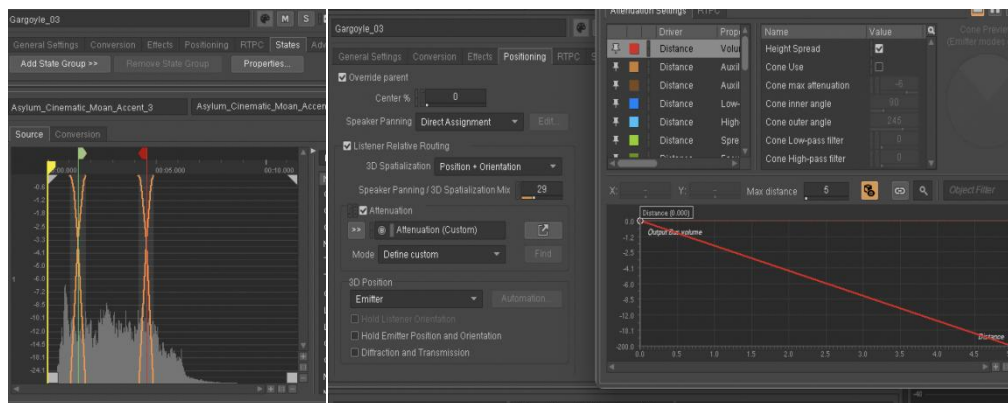


I designed SFX in Protools/Reaper, and composed my soundtrack Cubase. After that I imported them into my Wwise project. I designed several events that change the play mode of certain SFX, for example: I used a switch container for 3 different footsteps; I applied States to change the heartbeat sound to mimic a safe room environment, when John Lemon enters the safe area, the music adds a Low-pass filter, a reverb, the heartbeat also starts to fade in gradually.

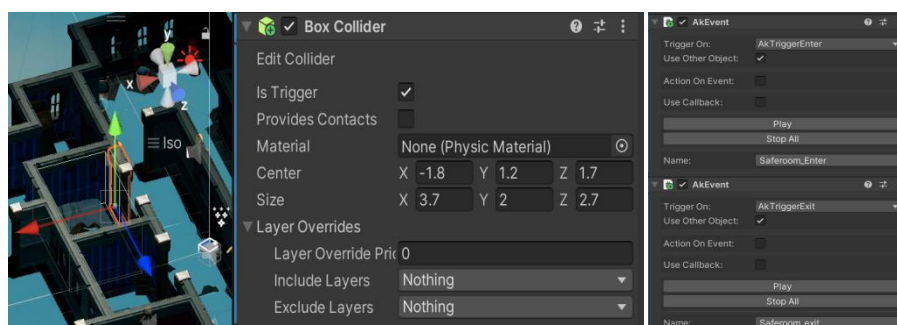


After that, I generated the soundbank and integrate my Wwise project into Unity for testing (since the game lacks voice for characters, I wasn't able implement voiceovers).

In Unity, I began to add sound the ghost sounds. I created two distinct samples for this, adjusted the Positioning parameters in Wwise based on the relative positions of the kitten and the ghost. For the gargoyle, I created three similar samples that could be played randomly and loops automatically.



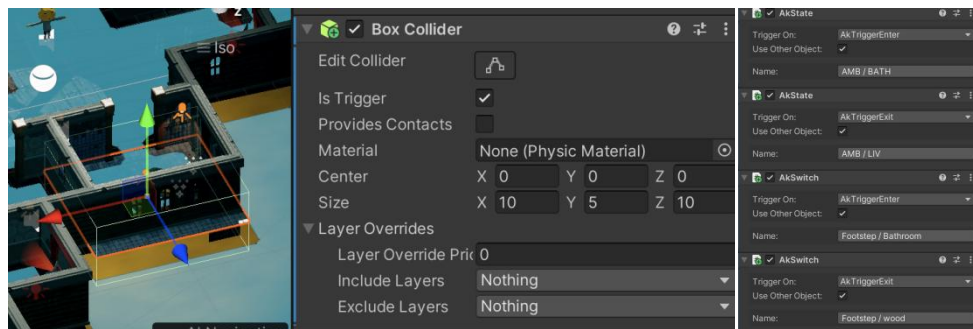
In terms of music, I used Cubase as my DAW, blending orchestral instruments and synthesizers to create a quirky yet eerie soundtrack. At the end of each music segment, I added short glissandi from the piccolo to introduce a playful touch. In Unity, I placed sound triggers in each room to transition the music's state when the kitten entered, simulating the kitten's sense of fear while amplifying the heartbeat.



Additionally, I designed sound effects for winning and losing. For the loss event, I

created a chilling sound accompanied by a descending melody from the music.

For environmental sounds, I designed 3 sfxs in game: rain from outside, a shower in the bathroom, and the flickering lightbulb in the hallway. The rain was present throughout the scene, while I placed a box collider in the bathroom to trigger the shower sound as the player approached. I also programmed automatic transitions between different footsteps based on the type of surface, such as tile or wood. Upon exiting the bathroom, the sound would return to the living room environment. I added two Akevent to lightbulb components and used Positioning to adjust the volume based on the kitten's proximity to the lightbulbs.



The most challenging aspect of this project was the mixing process. While I had set the loudness range between -23 and -18 LUFS, achieving a balanced sound experience within the game was still difficult. After completing the mix in Wwise, I connected it to Unity, allowing for real-time adjustments based on the in-game audio. During this phase, I consulted with my mentors from internships at Tencent and Chao Parametrics, who offered invaluable insights and advice. Their guidance on setting up the project and their feedback on creative approaches were extremely helpful, and I truly enjoyed collaborating with them.